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MILITARY SPECIFICATION

PRINTER, PROJECTION, PHOTOGRAPHIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable projection printers, complete with lens and accessories, for the production of enlarged prints from 35mm film negatives.

1.2 Classification. The printers shall be the following types as specified (see 6.2).

Type I. - Printer, Projection, Photographic EN-15( ) with baseboard.
Type II. - Printer, Projection, Photographic EN-11( ) without baseboard.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530 - Plywood, Flat Panel.
MMM-A-181 - Adhesive, Phenol, Resorcinol Or Melamine Base.

MILITARY

MIL-P-116 - Preservation, Methods Of.
MIL-C-675 - Coating of Glass Optical Elements (Anti-reflection).
MIL-C-3432 - Cable (Power And Special Purpose) And Wire, Electrical (300 And 600 Volts).
MIL-C-3767/13 - Connector, Plug, Electrical (Power, Three-wire, Fixed Grounding Type (Type UP221M).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6720

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MIL-P-4672 - Paper, Photographic, Black And White, Sensitized.
MIL-D-4827 - Developer, Photographic, Paper, Armed Forces Developer No. 25.
MIL-F-10336 - Fiber Sheet, Vulcanized.
MIL-F-14072 - Finishes For Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-130 - Identification Marking Of US Military Property.
MIL-STD-150 - Photographic Lenses.
MIL-STD-252 - Wired Equipment, Classification Of Visual And Mechanical Defects.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.

2.1.2 Other Government documents. The following Government documents form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

SC-A-19100 - Hardware for Chests and Cases.
SC-A-47794 - Wood Guide.
SPI IG00466 - Special Packaging Instructions.

(Copies of specifications, standards and drawings required by manufacturers in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The printer shall consist essentially of a lamphouse, a carriage, a condenser unit, a projection lens, a support structure, and when type I is specified, a baseboard. When using 35mm film, the printer shall be capable of making projection prints in sizes varying from 2.35 by 3.25 inches to 8.5 by 12.5 inches (57 by 83mm to 216 by 318mm).

3.2 First article. The quantity of samples specified in the contract shall be subjected to first article inspection (see 4.3 and 6.2c).

3.3 Material. The material shall be as specified herein and as shown on the applicable drawings. However, when a definite material is not specified, a material shall be used which will enable the printer to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled material. It is encouraged that recycled material be used, when practical, provided it meets the requirements of this specification.

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3.3.2 Cleaning. Material, parts, and equipment shall be cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, solder, flux residues, outgassing products and any other contaminant. The cleaning procedure and cleaning agents shall have no deleterious effect on the performance requirements of the materials, parts and equipments. The same cleaning procedure shall be applied to all printers produced and delivered. All cleaning of material, parts and equipment shall be in accordance with requirement 9 of MIL-STD-454.

3.4 Design.

3.4.1 Baseboard. When type I printer is specified, the baseboard shall be constructed of type I, grade I, birch hardwood plywood, approximately 0.8125 inch (20.6mm) thick. The plywood shall be sanded on the faces and edges, and finished with a clear lacquer. A means of attaching the support structure to the baseboard shall be provided.

3.4.2 Support structure. The printer shall be equipped with a support structure of rigid metal to permit vertical adjustment of the carriage assembly. The structure shall permit normal operation of the printer without any vibration detrimental to the making of prints. A spring type counterbalance shall be attached to the support structure which shall balance the carriage assembly at all times during its travel over the entire range of the support structure. The support structure shall position the carriage assembly over the longitudinal center of the baseboard.

3.4.2.1 Attachment. The bottom of the support structure shall permit easy attachment to and detachment from the baseboard without the use of tools. Such provisions shall also permit the support structure to be turned 180° in a horizontal plane to provide a capability for making projection prints at the floor level of the darkroom.

3.4.3 Carriage assembly. The lamphouse, condenser unit, and lens assembly shall be mounted on a carriage constructed of rigid metal. The carriage assembly shall provide smooth and accurate travel along the full length of the support structure without introducing any variation in alignment of the printer that would impair the making of prints. A friction locking device shall be provided for locking and unlocking the carriage at any point along the support structure. Provisions shall be made to rotate either the enlarger head or the 35mm film holder in a horizontal plane to facilitate composition.

3.4.4 Lamphouse. The lamphouse, constructed of cast aluminum, shall be self-ventilating and light-tight. It shall dissipate the heat of the lamp from other parts of the printer, and under normal operating conditions, the temperature of the negative shall conform to 3.7. The optical system of the lamphouse shall illuminate the surface of the negative uniformly. Provisions shall be made to permit easy access to the lower condenser face and film carrier for cleaning, and also to the lamphouse to install or change the projection lamp without the use of tools.

3.4.5 Condenser unit. A condenser system shall be provided having a combined 3.5 inch (89mm) focal length, and shall be in accordance with the applicable provisions of MIL-STD-150 for type XI lenses. The lenses shall be free from striae, occlusions, and other defects which will interfere with the quality of projection prints obtained with the printer. The condenser unit shall collect and concentrate the illumination uniformly over the maximum negative area. The condenser unit shall permit rapid and convenient attachment to and detachment from the lower part of the lamphouse.

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3.4.6 Film carrier. The film carrier shall accommodate 5.5 feet (1.7 meters) of 35mm film, and shall be constructed to permit the insertion of film without the removal of the carrrier between exposures. The carrier shall hold the film flat, centered in and perpendicular to the optical axis within the projection aperture during printing. The projection aperture shall have suitable dimensions for printing a 24mm by 36mm (double frame) format. Provision shall be provided to prevent the scratching of the film when being moved through the film carrier.

3.4.7 Electrical. The printer shall be capable of operating from the following power sources:

a. 115 volts AC ±10 percent, 60 Hertz ±5 percent.
b. 115 volts DC ±10 percent.
c. 6 volts DC.

3.4.7.1 Power cord.

3.4.7.1.1 Type I. A 10 ±.25 foot (305 ±8cm) flexible power cord, consisting of three #18 AWG standed wires in accordance with MIL-C-3432, shall be permanently attached to the printer. It shall terminate in a 3-wire, grounding type, electrical connector plug Type UP221M in accordance with MIL-C-3767/13. The cord shall be provided with a feed-through ON and OFF snap-type switch, positioned so that it is convenient to the operator.

3.4.7.1.2 Type II. A 48 ± 1 inch (122 ± 2.5cm) flexible power cord, consisting of three #18 AWG stranded wires in accordance with MIL-C-3432 shall be permanently attached to the printer. It shall terminate in a 3-wire plug conforming to MIL-C-3767/13, type UP221M.

3.4.7.2 Dielectric strength. Following the exposure to the moisture resistance test (see 4.6.7.3), there shall be no flashover or breakdown when a potential of rated voltage plus 1000 volts is applied between any electrical circuit and ground or between any electrical circuit and adjacent metal parts not in the circuit under test.

3.4.8 Dimensions and weight.

3.4.8.1 Type I printer. The printer, when set up for operaton, shall be not more than 36 inches (91cm) high, 20 inches (51cm) wide, and 30 inches (76cm) deep. The weight of the complete printer without carrying case, shall be not more than 25 pounds (11.3kg).

3.4.8.2 Type II printer. The printer shall be not more than 20.5 inches (52cm) high, 5 inches (13cm) wide, and 13 inches (33cm) deep when in a non-operating position. The weight of the printer shall be not more than 18 pounds (8.2kg).

3.5 Optical system operation.

3.5.1 Projection lens. A nominal lens of approximately 50mm (2-inch) f/4.5, type VI lens in accordance with MIL-STD-150, shall be provided with the printer. The lens shall have diaphragm openings of f/16 (min.) and f/4.5 (max.). The diaphragm of the lens shall be actuated by a ring having a positive click stop for each aperture from minimum to maximum. The mounting method shall ensure that the lens is centered in the optical

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axis of the printer and maintained with its axis perpendicular to the film. The effective focal length of the lens shall be within ±4 percent of the marked value. The images produced by the lens used shall be sensibly free from flare or spurious images due to internal barrel reflections, surfaces, or glass defects. The lens shall be coated in accordance with MIL-C-675 and shall be furnished with a lens cap.

3.5.2 Focusing assembly. A manual focusing device shall be included in the carriage assembly to provide for fine focusing of the projection lens throughout the printing range.

3.5.3 Filter kit. The printer shall be capable of accepting the filter holder and the polycontrast filters in Kodak Filter Kit (Model A) or equal.

3.5.4 Distortion. When a test plate is projected, the projected image of the lines adjacent to both the long and short edges of the picture aperture shall depart not more than 1.0 percent of their length from a straight line joining opposite ends of the projected image of the line.

3.5.5 Resolving power. The minimum resolving power of the lens, with the diaphragm set at its largest opening, shall be not less than 56 lines/mm at the center, and not less than 40 lines/mm at any other point in field.

3.5.6 Uniformity of illumination. At maximum magnification, the ratio of illumination of any corner to that of the center of the projected area shall be 60 percent or greater, and the field shall be free from local irregularities having intensity differences of more than 15 percent of the illumination adjacent to the irregularities. Illumination of the projected area shall also be free from the objectional bands or patches differing in color or brightness from the adjacent parts of the illuminated area.

3.5.7 Printing speed. The printing speed shall be such that when a neutral density filter of 1.00 is exposed at f/4.5 for 16 seconds at 4 times magnification onto paper conforming to MIL-P-4672, type II, subtype A, contrast 2, the resulting print density will be not less than 0.63.

3.6 Environmental conditions. The printer shall withstand individual environmental conditions specified herein, or combinations thereof, without damage, performance degradation, or malfunction when tested in accordance with 4.6.7.

3.6.1 Operational.

a. Temperature range: 62°F (16.7°C) to 110°F (43°C).
b. Relative humidity up to 95 percent.

3.6.2 Non-operational and storage.

a. Temperature range: 160°F (71°C) to -60°F (-51°C). Exposure time not more than 24 hours per each extreme at any one time.
b. Relative humidity up to 95 percent.
c. Vibration: No mechanical resonance below 55 hertz.
d. Stability: 20° tilt in any direction.

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3.7 Temperature rise. The maximum temperature of the surface of the test film shall not exceed 122°F (50°C) after the printer has been in intermittent operation for a period of 60 minutes in an ambient temperature of 70°F (21.1°C).

3.8 Carrying case. When specified in the contract (see 6.2), a dustproof and splashproof carrying case shall be provided for type I printers. The case shall be constructed of plywood and fiber.

3.8.1 Plywood. The plywood shall be 0.375 inch (9.5mm) thick Douglas fir or hardwood plywood in accordance with NN-P-530 and Drawing SC-A-47794.

3.8.2 Fiber. The fiber used shall conform to MIL-F-10336.

3.8.3 Adhesive. Laminating of the fiber to plywood shall be accomplished using adhesive conforming to MMM-A-181.

3.8.4 Hardware. The hardware for the case shall be as specified in Drawing SC-A-19100.

3.8.5 Storage compartments. The storage compartments shall be provided to hold all components of the printer.

3.8.6 Dimensions and weight. The carrying case shall be not more than 48 inches (122cm) long, 21 inches (53cm) wide, and 9 inches (23cm) high. The case, without the printer or components, shall weigh not more than 25 pounds (11.3kg).

3.9 Paper holder. When specified, (see 6.2), a photographic paper holder shall be provided.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.11 Finish. The printer shall be given protective finish in accordance with MIL-F-14072, and as specified on the drawings.

3.12 Marking. Marking shall conform to MIL-STD-130.

3.12.1 Serial numbers. Each printer shall be serial numbered.

3.13 Workmanship. The printer shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspecton requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections coverd by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When specified in the contract, the contractor shall furnish first article units of the printer (see 6.3).

4.3.2 First article inspection. This inspection shall consist of the inspection specified in the subsidary documents covering items listed in 4.4 and the inspection specified for group A, group B, and group C (see tables II, III, and IV, respectively). The first article inspection shall be performed in the order shown in table I according to the number of samples specified in the contract. After completion of all group C tests, conforming units shall be reinspected and shall pass all group A tests. Failure of any inspection shall be cause for rejection.

Table I. Order of first article inspection.

Inspection 1/	Order of inspection		
	One sample	Two samples	
		Unit 1	Unit 2
Non-damaging: 2/			
Low temperature	1	1	
High temperature	2	2	
Vibration	3		1
Potentially damaging:			
Humidity (moisture resistance)	4		2

1/ Other first article inspection may precede, follow, or be interspersed between the non-damaging tests.

2/ The order of non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration test shall be performed on the same unit used for the bounce and shock test.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements.

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Item	Where required
Condenser unit	3.4.5
Power cord	3.4.7.1
Projection lens	3.5.1
Lens coating	3.5.1
Plywood	3.8.1
Fiber	3.8.2
Adhesive	3.8.3
Finish	3.11
Marking	3.12

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibilities for performing any additional inspection which is necessary to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit shall be subjected to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined, or as specified herein, shall be in accordance with MIL-STD-105. Lot sampling shall be in accordance with MIL-STD-105. Acceptable quality level (AQL) shall be as specified in table II.

Table II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical	3.13	4.6.10	1.0	4.0
Operational				
Distortion	3.5.4	4.6.3	1.0	
Resolving power	3.5.5	4.6.4	1.0	1/
Uniformity of illumination	3.5.6	4.6.5	1.0	
Printing speed	3.5.7	4.6.6	1.0	

1/ All operational defects are considered major.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III.

Table III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Interchangeability	3.10	4.6.9	6.5

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4.5.2.1 Group B sampling. Sampling for group B shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be as specified in table III.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in any order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV.

Table IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Order of inspection
Dielectric strength	3.4.7.2	4.6.1	1
Temperature	3.6.1a, 3.6.2a	4.6.7.1, 4.6.7.2	2
Humidity, moisture resistance	3.6.1b, 3.6.2b	4.6.7.3	3
Vibration	3.6.2c	4.6.7.4	4
Stability	3.6.2d	4.6.7.5	5
Temperature rise	3.7	4.6.8	6

4.5.3.1 Sampling for Group C inspection. One sample unit shall be selected at random for inspection from every 50 units, or fraction thereof, produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed as specified in table IV.

4.5.3.3 Group C failures. Action required relative to Group C failures shall be as specified (see 6.2).

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are subjected to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Dielectric strength. A test potential of rated voltage plus 1000 volts at 60 Hz AC, or DC of the same value, shall be applied for one minute between all wiring and adjacent conductors not in the circuit and between all wiring and ground. Nonconformance to 3.4.7.2 shall constitute failure of this test.

4.6.2 Operational. The printer, set up for operation, shall be tested as follows:

a. The carriage shall be traversed the full length of the support column 5 times to ensure ease in opeation.

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b. Projection prints shall be made from 35mm film with the carriage at maximum and minimum positions. The resulting prints, processed as specified in 4.6.6, shall be examined for photographic quality.

c. Place the printer on photographic projection paper covering a square area of 3 by 3 feet (91 by 91cm). Cover the baseboard with a sheet of projection printing paper. Place the lens cap on the lens and turn on the projection lamp for a period of 2 minutes. Thoroughly process the photographic paper used in this test. Compare the processed paper that was covered by the baseboard. Any evidence of exposure to light shall constitute failure of this test.

4.6.3 Distortion. With the printer set up for projection on the floor, the image of a test plate having straight lines adjacent to the four edges of the film aperture, shall be projected at full magnification on a smooth white paper surface. The projected image of the straight lines shall be compared with straight lines joining opposite ends of the projected image of the lines on the white paper and shall not be distorted more than specified in 3.5.4. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6.4 Resolving power. The resolving power of the projection lens shall be as specified in MIL-STD-150, method 14. Nonconformance to 3.5.5 shall constitute failure of this test.

4.6.5 Uniformity of illumination. The printer shall be set at its maximum enlargement ratio, and the light beam projected through the glass holder masked for its maximum film size. A foot-candle meter or other photometer shall be used to determine the uniformity of illumination. Measurements shall be made at the center and four corners of the illuminated area. The illumination at any corner of the image plane shall be not less than 60 percent of the illumination on axis. The presence of objectionable bands or patches differing in color or brightness from the adjacent parts of the illuminated area shall constitute failure of this test.

4.6.6 Printing speed.

4.6.6.1 Densitometer. The densitometer used to determine the transmission density of the neutral filter and the reflection density of the print from the filter shall be of the Ansco-MacBeth type, or equal.

4.6.6.2 Densitometer adjustment. The instrument shall read zero when a piece of photographic paper, not developed but fixed out, is placed in the instrument and a reflection density reading taken. If the instrument does not read zero, the reading shall be subtracted from the reading of the test prints. The zero reading shall be taken before each set of test prints is taken, and shall be used only for that set of test prints.

4.6.6.3 Procedure. With the lens aperture set at f/4.5 and the printer set for 4 times magnification, a neutral density filter of 1.00 $\pm$0.05 shall be projected onto photographic paper in accordance with MIL-P-4672, type II, subtype A, contrast 2 for a period of 16 seconds. The paper shall be processed for 1.5 minutes in Armed Forces Developer No. 25, in accordance with MIL-D-4827, diluted 1 part developer to 2 parts water, at 68 $\pm$2°F (20 $\pm$1°C). The paper shall be air dried to a semi-matte surface. The print density shall be measured with a densitometer. Nonconformance to 3.5.7 shall constitute failure of this test.

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4.6.7 Environment.

4.6.7.1 High temperature. The printer shall be tested in accordance with MIL-STD-810, method 501.1, procedure II. The non-operating temperature shall be 160°F (71.1°C) for not more than 24 hours and the operating temperature shall be 110°F (43.3C).

4.6.7.2 Low temperature. The printer shall be tested in accordance with MIL-STD-810, method 502.1, procedure I. The non-operating temperature shall be -60°F (-51°C) for not more than 24 hours and the operating temperature shall be 62°F (16.7°C).

4.6.7.3 Humidity. The printer shall be tested in accordance with MIL-STD-810, method 507.1, procedure II. Prior to testing, the printer shall meet full performance for the measurements specified in 4.6.2. No repair or replacement of parts shall be permitted.

4.6.7.4 Vibration. The printer shall be subjected to sinusoidal vibration having a double amplitude of 0.76mm (0.030 inches), the frequency being varied uniformly between the limits of 10Hz and 55Hz. The rate of frequency change shall be linear and shall be such that a complete cycle (10 to 55 to 10Hz) shall be traversed in 30 minutes. One cycle shall be performed along each of the three major axes of the printer for a total test time of 90 minutes.

4.6.7.5 Stability. With a printer set up for operation, the carriage assembly set at maximum magnification shall be tilted 20° from the vertical in each direction parallel to the base.

4.6.7.6 Failure criteria. Nonconformance to 3.6 shall constitute failure of these tests.

4.6.8 Temperature rise. A test film having a uniform density of 2.0 or greater with a suitable thermocouple of 0.00396 or 0.00353 inch (0.1mm or 0.09mm) diameter (No. 38 or 39 AWG) wire held or cemented to the emulsion side of the test film at the center of the aperture shall be placed in the film holder. The printer then shall be operated intermittently for a period of 60 minutes, cycled 10 seconds ON and 10 seconds OFF. At the end of this period the temperature on the face of the negative shall be determined with an instrument accurate to $\pm$4°F (2.2°C). The test film shall be placed in the film carrier with the emulsion side away from the light source. Nonconformance to 3.7 shall constitute failure of this test.

4.6.9 Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of part.

b. Dimensions of cavities, when such dimensions affect insertions of items.

c. Location of fasteners on separable parts or assemblies which must mate, such as condenser unit, film carrier, filter holder, etc.

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4.6.10 Visual and mechanical inspection. The printer shall be examined for the defects listed in MIL-STD-252. Nonconformance to 3.13 shall constitute failure of this test.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table V, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent.

Table V. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

5. PACKAGING.

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI IG00466.

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6. NOTES

6.1 Intended use. The printers are intended for use in making projection prints from 35mm perforated or unperforated film. Type II printer is intended to be used with carrying case, Photographic Equipment FM-22 ( ) which is capable of housing the complete equipment assembly of a photographic set.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type printer required (see 1.2).
c. When a first article is required for inspection and approval, and number of units required (see 3.2).
d. When a carrying case is required (see 3.8).
e. When Holder, Photographic paper is required (see 3.9).
f. Group C failure action (see 4.5.3.3).
g. Level A or B preservation and packing (see section 5).
h. When first article inspection rough handling tests are not required.
i. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required the item will be tested and should be a first article. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of first article.

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: EN-15W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.2).

6.5 Carrying case. When the procurement document specifies that the printer be supplied without the carrying case, a simulated carrying case conforming to the mechanical requirements of 3.8 shall be furnished by the contractor so that the printer can be subjected to the vibration test.

6.6 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - AS
Review activities:
DLS - GS
User activities:
Army - MI
Navy - MC

Preparing activity:
Army - ER

(Project 6740-0140)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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